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Intellectual property rights and control in the digital economy: Examining the expansion of M-Pesa

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ABSTRACT

The importance of intellectual capital in the digital economy implies an increasingly central role of intellectual property rights (IPR). However, there are concerns that the expansion of intellectual property rights concentrates profits from innovation, creating “intellectual monopolies”. This is particularly concerning with respect to strong global regimes of intellectual property that foster economic inequities between the Global South and North. Such arguments imply that IPR is a core driver of structural inequality in the digital economy, yet there is very little empirical analysis of how such conceptual ideas unfold. This study focuses on IPR in the Kenyan mobile money service M-Pesa. It charts how M-Pesa expanded from a development-orientated innovation in Kenya to become part of a global enterprise. Control of IPR has led to bottlenecks in innovation in Kenya and significant South-to-North financial transfers. Overall, this case contributes to expanding the debate about patterns of innovation and governance of the digital economy. As digital firms expand, unpacking the processes by which global intellectual property regimes and cross-border IPR practices shape uneven power relations and inequality is vital.

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Introduction

There has been an emergence of global digital innovations that follow common patterns – operating at a global scale, owned by large firms, and shaped by demands of financialization, market dominance, information control, and data ownership. The idea of global-scale innovation driven by multinational firms is nothing new, but there is a sense that the dynamics and concentration in the digital economy may be more challenging (Srnicsek 2016).

To theorize such dynamics, scholars have looked to leverage the concept of rent: the ways that profits and control in the digital economy have been centered on “income derived from the ownership, possession, or control of scarce assets and under conditions of limited or no competition” (Christophers 2019, 308). New forms of rent creation are varied in the digital economy but have been discussed in terms of control of digital platforms, the enclaving of digital ecosystems, the use and control of data, investments in infrastructure, financialization, and control of

intellectual capital built by such firms (Arvidsson and Colleoni 2012; Birch and Cochrane 2022; Rigi and Prey 2015; Srnicsek 2016).

It is this latter formulation of rent associated with *intellectual capital* that is the focus of this article and its relationship to *intellectual property rights* (IPR). Intellectual property rights (IPR) include a range of different rights that protect individuals and organizations who create *intellectual property* (IP). Significant IPR includes copyright for creative works and computer programs, trademarks for brands and marks, and various rights on innovative ideas including patents, industrial designs and trade secrets. At a firm level, IPR has been positioned to be core to business models in the digital economy. This is because innovation and competitive advantage are increasingly defined by intangible capital which requires the protection of IPR (Haskel and Westlake 2017). At a global level, IPR has been discussed in terms of the structuring of uneven global core-periphery relationships (Chang 2001). With strong global intellectual property regimes strengthening the status of powerful IPR

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holders, expansions of IPR in the digital economy may further cement structural inequality.

Such arguments, at a firm and macro level, are well rehearsed but there are very few cases to unpack these processes in the digital economy. These dynamics should particularly be considered related to the seemingly contrasting innovation literature which highlights innovation emerging from increasingly diverse forms, locations and sources (Chesbrough 2006; Foster and Heeks 2013b). This leads to research questions about the realities of IP and IPR: Does the global expansion of digital innovation cohere with ideas of rent creation through exploiting intellectual capital (and the use of IPR)? If so, what are the dynamics of IPR such processes?

To address these questions, the article examines IPR in the case of the Kenyan mobile money service M-Pesa. It discusses how M-Pesa intellectual property has been fraught with tensions during its entire history. These issues have rarely been systematically discussed in the case, with a tendency to focus on the direct developmental benefits of this service, particularly the impacts of its adoption in the late 2000s and 2010s (e.g. Suri and Jack 2016). More recent literature has reconsidered M-Pesa within patterns of political economy and the capture of value which has cast a more critical light on this service (e.g. Bateman, Duvendack, and Loubere 2019; Tyce 2020). Such work suggests that M-Pesa should be analyzed as part of a network of actors with different agendas and complex flows of knowledge and control in the digital economy.

The remainder of the article is set out as follows. In the next section, it expands on three core ideas around IPR – global IPR governance, intellectual capital in the digital economy, and the expansion of innovation theories. In subsequent sections, it examines M-Pesa. It introduces M-Pesa and how it has become part of a broader corporate structure. It then discusses the key aspects of its IPR and its dynamics over time. In the last section, based on this analysis, it reflects more broadly on the implications of this case. Overall, the analysis emphasizes the globalization of this service, with an associated expansion of IPR. Intellectual property is a key source of control of innovation, and ownership has been skewed toward more powerful actors within the corporate network. Not only has this led to significant bottlenecks in innovation in Kenya, but also a source of significant financial transfers which call into question more optimistic claims of development associated with this service.

IPR and digital innovations

Beginning from the notion of rent, one might roughly identify two contrasting positions that have been taken in the literature related to intellectual capital and IPR (Foster 2023; Morozov 2022). Work stemming from Richardian ideas of rent, see rent linked to unearned income and inequality, whereby IPR becomes part of a story of market distortion (Foley 2013; Mazzucato 2018). In contrast, the literature on innovation and IPR more closely sits within Schumpeterian analysis where short-term rents emerge from innovative activities and associated productive dynamics (Baglioni, Campling, and Hanlon 2021; Rikap 2022).

To translate these two ideas into a more concrete analysis, three major debates on IPR are reviewed. Firstly, the global institutions of IPR, and how they shape structural inequalities. Secondly, the emergence of the digital economy and its expansionary and monopolistic character. Thirdly, diversity and changing norms of digital innovation suggest contrasting patterns around the creation of IP.

Structural inequalities

The notion of intellectual property (IP) emerged in the 16th-18th centuries as firms began to look to protect products and machines from being copied (Peukert 2017). Poignantly for the topic of this article, one of the key drivers to this was the expansion of European colonialism where intellectual property rules allowed firms to more profitably expand across their colonies (Kongolo 2013; Peukert 2012). Over the 19th and early 20th centuries, ad-hoc regimes were increasingly formalized nationally and across borders. Following the initial Paris Convention in 1883, and then adapted throughout the early twentieth century, intellectual property was increasingly legislated for at a super-national level (Chang 2001). Core paradigms of IPR established in these earlier regimes still exist today, for example, the “right of priority” that enables IPR owners in one country to have rights across other countries party to IP agreements (Chang 2001). Moving forward to recent decades, major developments in IPR came with the emergence of dedicated and powerful global intellectual property institutions. The establishment of the World Intellectual Property Organization (WIPO) and the ratification of the article on Trade-Related Aspects of Intellectual Property Rights (TRIPS) in 1995 within the World Trade Organization (WTO) has led to the establishment of strong intellectual property regimes across the members of the WTO (Peukert 2017).

Debates on the value of IPR were particularly intensive during the negotiations and early implementation of TRIPS. This was the case because TRIPS led to strongly opposing positions about how intellectual property related to structural inequality. Key advocates (particularly led by corporations and technologically advanced nations) argued that strong, global IPR regimes supported foreign investment and enabled the expansion of domestic innovation through enhanced protection (Ginarte and Park 1997). More critical scholars highlighted that strict intellectual property rules could limit the transferability of knowledge that comes through global technology diffusion and learning. Strong regimes of IPR would also narrow the ability of governments to shape domestic innovation through industrial policy (Chang 2001). IPR predominantly supports multinationals who can create and litigate within a global infrastructure of IPR at the expense of domestic innovators (Markusen 2001). Nearly three decades after the introduction of TRIPS, much of the evidence of global IPR regimes suggests overall negative consequences (Chang 2001; Peukert 2017; Stiglitz 2014). There can be potential benefits, as seen in the growth of regional innovators in some emerging markets, but these more optimistic outcomes appear to be quite limited (Ramamurti 2004).

At present, even marginal reform to TRIPS is being resisted by powerful nations. Nevertheless, it is useful to highlight recent debates that push for IPR regimes to be more inclusive. An important debate relates to how social- and public-orientated research institutions can gain from IPR (nonprofit, public, universities and social enterprises). Such actors often poorly fit with IPR regimes which tend to conceptualize IPR creators as individual “inventors”. Moreover, public organizations that are involved in innovation may have agendas that move beyond profit motives. There has been an effort to rebalance these limitations, for example, countries such as the US and UK have pushed strategies to empower public R&D institutions to more actively reap intellectual property rewards (Montobbio 2009). There have also been some attempts to mirror these strategies globally (Graff 2007). However, such approaches have limits, particularly when profit and social agendas clash. This has been vividly illustrated in recent debates in the global pharmaceutical industry and vaccines (Ghauri and Rao 2009). Even with large-scale public funding and pressure, control of IPR has led to profits accrued to commercial partners in the global North with uneven consequences.

Monopolization in the digital economy?

A well-established idea underlying IPR regimes is that they incentivise innovation by granting a “temporary monopoly” to creators of IP. This temporary monopoly is seen as a beneficial tradeoff in terms of overall societal gains. However, this is a delicate balance. With IPR regimes expanding there are growing questions if regimes are moving too far toward monopolization (Hall and Harhoff 2012). This is illustrated, for example, in the significant expansion in IPR activity (Zukerfeld 2017). WIPO data shows a growth in global patents from 1.6m filings in 2004 to 3.3m in 2018, and trademarks applications from 4.6m in 2004 to 14.3m in 2018 (WIPO 2020).

A key driver of this trend has been the importance of IPR to support intangible capital, related to technological innovation. This position has been emphasized in discussions on the digital economy. Intangibles, investments made in intellectual assets, are increasingly important in an economy driven by information (Haskel and Westlake 2017). Digital firms, in particular, make a strong focus on intellectual capital over other forms of capital. Therefore, IPR provides a way to protect these investments and is foundational to a firm’s fortunes, investment patterns and directions (Benkler 2006; Shapiro 2000).

An important parallel trend is that national IPR institutions are allowing a broader set of IP to be protected under IPR and granted for increasing periods (Benkler 2006). Such expansion is particularly driven by IPR institutions in certain countries which have been more lenient in granting patents (such as the US) (Shapiro 2000). In the digital economy, the challenge of these expansionary dynamics of IPR has been illustrated in the case of “software patents” (Benkler 2006; Lessig 2004; Stallman 2002). Software has long been a challenge in terms of IPR regimes. Software code is covered under copyright, but this may not protect the inventive aspect of the processes encompassed by code (Nichols 1999). Patents have historically excluded software, with a focus on novelty as non-obvious changes in devices, machines, and tools (EPO 2020). Novelty in software was seen as an abstract idea and outside of scope. While still a much debated area, over time national IPR agencies have loosened their rules on software patents and this has been an important part of facilitating the expansion of IPR (Bessen and Hunt 2007; Hall and Harhoff 2012).

Overall, these dual drivers – the expanding role of IPR in the digital economy and the broader range of novelty covered by IPR – are significant. Technology-intensive firms are increasingly building

large interrelated IPR portfolios. More critical scholars have emphasized that such expansion of IPR is central to forms of monopoly and control by those who control IPR (Foley 2013). In addition, it leads to the appropriation and monopolization of public knowledge and practices by commercial actors (Rigi and Prey 2015). The economic implications of these dynamics are that monopolies can begin to limit the types of ideas, processes, and innovations that remain within “the commons” leading to skewed motivations for innovation in society (Stiglitz 2014).

More critical accounts of these trends within the digital economy would go further. Shaped by IPR regimes (and other intellectual capital such as data) they argue that the most crucial aspect of economic rent should now be seen in the domain of intellectual capital. “Intellectual monopoly capitalism” is leading to the wholesale privatization and exploitation of the commons by global intellectual monopolies, that move beyond market control (Pagano 2014). As the link between IPR portfolios and the actual implementation of innovations becomes tenuous, IPR becomes an important part of firm assets and valuation in a financialised economy (Rice 2015). These dynamics have been particularly important in the tech sector with its huge valuations and capital investment. Building IPR assets then indicates an ability to create future rents and is central to present and future market valuation (Birch 2020).

There has been limited success in pushing back against expansionary IPR trends, including some successes in campaigns against software patents in certain regions. In terms of development, an important global debate has also been proposals to limit the enclosure of the commons through IPR related to local and indigenous knowledge, creativity, and innovation. Such discussions have stemmed from high-profile cases when indigenous knowledge has been commercialized by corporate actors including IPR with significant profits outside their origins. Key areas include biopiracy (e.g. plants, foods), traditional knowledge (e.g. techniques, remedies) and culture (e.g. crafts, music) (Schuler and Finger 2004). Attempts to reduce such activity look to protocols around IPR related to indigenous knowledge and folklore, as well as integrated into frameworks on biological diversity (Kraemer-Mbula and Wamae 2010).

Innovation

Given that the objective of IPR is to incentivize innovation, it is appropriate to consider the changing patterns of innovation as related to the digital economy. An importing starting point is a now well-established

idea that innovation is no longer the preserve of only the most advanced actors, or just those involved in advanced R&D. Moving beyond linear models of innovation, more heterogeneous ideas of innovation such as systems of innovation, open innovation, and user-driven innovation center the broader contexts and multiple contributors involved in innovative processes (Chesbrough 2006; Lundvall 1992). In the context of the Global South, this implies a broad set of local actors involved in incremental activities that should be considered within studies of innovation (Foster and Heeks 2013b).

Over the previous decade, there has been a wealth of digital innovations that have been employed incorporating development goals and intentions around inclusion (followed by paradigms linked to ICT for development, ICT4D). Such cases have highlighted the involvement of diverse actors including governments, NGOs, and the active role of communities (Murphy and Carmody 2015). While many pilot projects did not last beyond their short periods of funding, there were successful interventions that overcame technology challenges and were able to operate sustainably and begin to scale. With the maturing of such innovations, scaled products and practices often become aligned more globally with actors involved in the digital economy (Foster et al. 2022; Graham 2018).

In sum, bringing the directions of the review together, perspectives on the digital economy center ideas of diffuse and diverse innovation. With the global expansion of the digital economy, this implies a far broader set of actors and contributions toward innovation. An important area to supplement understanding then is if, and how, heterogeneous digital innovation coheres with the expansionary IPR regimes. If the digital economy is marked by concentrating and increasingly monopolistic IPR, how are diverse innovations controlled and regulated within IPR regimes? Some work has begun to provide insight into this, focussing on the activities of multinational firms in the digital economy. By coordinating digital ecosystems, or data infrastructure, multinational firms may be able to control increasingly heterogeneous networks involved in innovation. Through this control, they can appropriate intellectual capital (Durand and Milberg 2020). Notions of global corporate innovation and production systems suggest similar directions that position leading firms focussed on systems for the institutionalization of knowledge that can be used for their own ends (Rikap 2022; Rikap and Lundvall 2022). Nevertheless, there remains a need for further analysis of these relationships, particularly linked to innovation emerging outside technologically leading nations.

M-Pesa and intellectual property

Approach

This article uses the single case of the mobile money service M-Pesa to begin to unpack in more detail some of these questions around the ambiguities of IPR and the digital economy. The goal is to provide rich descriptions of the activity around IPR, drawing on empirical data as the basis for improved analytical understanding and practice (Eisenhardt 1989). The case is of special interest as a highly successful innovation, first introduced in Kenya and then scaled over time. Moreover, having been extensively researched, it provides a richer set of empirical data to consider the nexus of innovation and IPR. In effect, it represents a *critical case*, that builds a detailed understanding of evolving IPR in an industry (Miles and Huberman 1994). As the in-depth account illustrates, many aspects in this case are context-specific, and thus the goal is not to generalize from this single case. However, detailed cases are needed to provide clearer explanations of IPR and digital innovation. Going in-depth into the intricacies of a single case is particularly important for this topic, where IPR is entwined with the complexity of firm relations.

The work is based on a long-run qualitative analysis of M-Pesa undertaken by the author. Data come from two types of sources. Firstly, the discussion builds on interviews undertaken in Kenya during two periods: 2010–2012 and 2015–2016 with staff, agents, and customers of the service –providing the basis for understanding the M-Pesa’s evolution. Alongside this, the author has been involved in conversations with various policy and practice actors¹. Secondly, to support this discussion and to dig into the IPR, it analyses a range of technical documentation on IPR including patents, project evaluations, and company filings. These are vital to illuminate the often technocratic aspects of IPR.

The next section provides an overview of the case, positioning M-Pesa as a global innovation, and summarizing the IPR. The subsequent sections then outline a roughly chronological account of how intellectual property has been generated and used in the service.

Positioning M-Pesa

Introduction to M-Pesa

M-Pesa is a Kenyan mobile service that allows money to be exchanged between mobile phone users. The

initial novelty of M-Pesa was that it was simple enough to be embedded within a mobile SIM card. This allowed the service to be used by a broad range of devices, including simple phones. Using the mobile number, a sender can transfer mobile money to a recipient using mobile messaging. Users who register with the service can, therefore, send or receive digital cash through their mobile number.

M-Pesa was initially a convenient mechanism for person-to-person transfers of money and this was the basis of its success in Kenya. These types of transfers remain an important component of the service today, particularly for low-income groups. Over time, M-Pesa has been incorporated into a range of payment services including within stores, banking services, insurance, savings, business-to-business payments, and government transfers (as outlined in the statistics in Table 1).

The Kenyan mobile operator Safaricom runs the operations of M-Pesa in Kenya, with the support of over 200,000 independent mobile agents (who supply “cash-in”/“cash-out” services) and with a growing ecosystem of other actors as the service expands. As outlined in the table, there are competitors in Kenya but M-Pesa has essentially become a monopoly.

Given the novelty and rapid scaling of M-Pesa, there has been a detailed discussion of the birth and rapid growth of M-Pesa, concentrating on the patterns of service management (For detailed accounts, see Foster and Heeks 2013a; Hughes and Lonie 2007; Ngugi, Pelowski, and Ogembo 2010; Omwansa 2009). The foundations of the M-Pesa service emerged in a pilot project undertaken in 2005. This was a project originating from the UK’s Department of International Development (DFID) in partnership with UK-based Vodafone Group (Safaricom’s parent company). Following the pilot, the service was introduced by Safaricom as a commercial service with rapid adoption and scaling. Over time, M-Pesa revenue (mainly generated through commissions in the service) has become an increasingly important aspect of Safaricom’s financial health as outlined in Figure 1. In 2008 it represented 4.2% of revenue, expanding to 32.1% by 2019. With the saturation of mobile users, alongside internet data, M-Pesa has become a driver of firm profitability.

With the expansion of the service, scholarship on M-Pesa has looked to align the service with discussions of financial inclusion and fintech for development, notably through statistical surveys. These have argued that mobile money use is having significant development effects in Kenya (Suri and Jack 2016; World Bank 2011). More critical discussions of rent and the political

Table 1. Mobile money statistics in Kenya, Q3 2019–2020 (January–March 2020).

	Total for Kenya	M-Pesa	M-Pesa proportion of total
Active mobile money subscriptions	29,185,577	28,842,584	98.8%
# Mobile money agents (local, independent cash-in, cash-out providers)	202,102	173,259	85.7%
Mobile money transfer categories			
Business-to-business (e.g. payments between registered businesses)	\$8,360,196,019	\$8,360,196,019	100.0%
Person-to-person (e.g. individual cash transfers)	\$6,417,217,788	\$6,413,246,371	99.9%
Business-to-consumer (e.g. salary payments)	\$3,477,971,699	\$3,475,750,580	99.9%
Consumer-to-business (e.g. retail payments)	\$2,955,776,655	\$2,952,232,096	99.9%
Consumer-to-government (e.g. tax, public utility payments)	\$221,914,717	\$108,827,318	49.0%
Mobile money deposits (e.g. users converting cash to mobile money)	\$5,789,785,352	\$5,774,508,633	99.7%

Source: CAK (2020) based on operator returns.

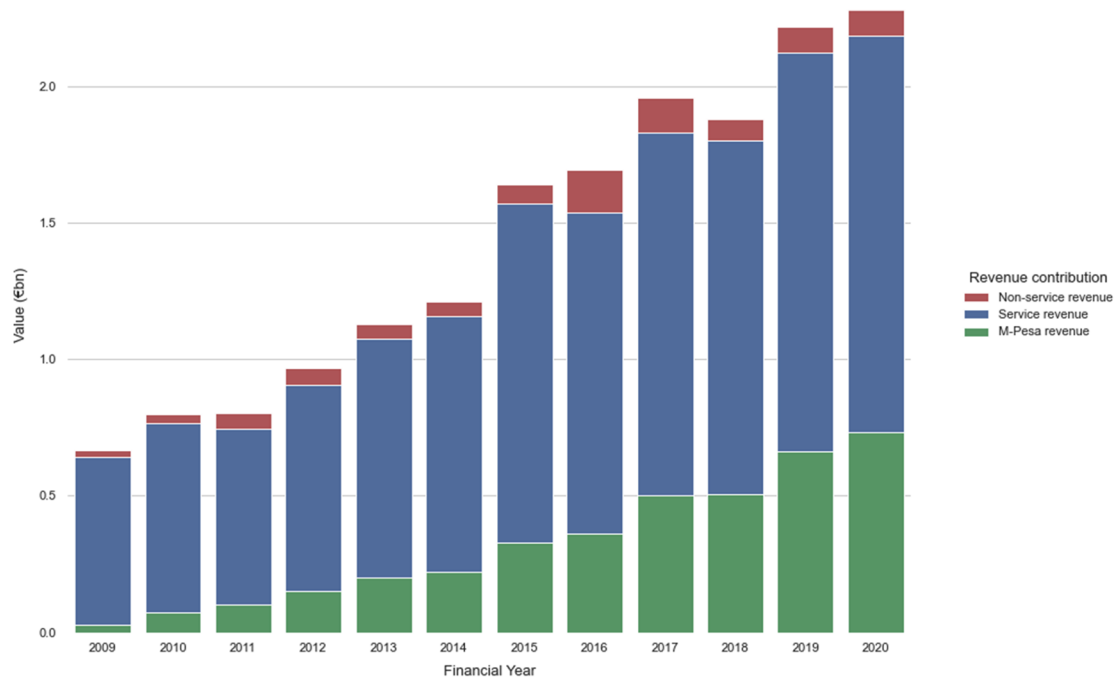


Figure 1. M-Pesa revenue. (a) Service revenue denotes revenue from operator services (voice, messaging, mobile data, fixed lines, excluding M-Pesa). (b) Non-service revenue denotes other revenue, primarily the sales of mobile handsets.

Source: Authors calculations based on Safaricom financial reports, € conversion based on respective the end-of-year rate.

economy of the service have been less prominent but recent work has begun to provide such discussion. Aligning with the growing critiques of development claims around fintech, observations have been made that the service has mainly benefited those controlling and developing it (Bateman, Duvendack, and Loubere 2019; Donovan and Park 2019). Other work has begun to unpack the intricacies of politics and labor, contrasting the lives of implementing agents and the elite political settlements related to the service within Kenya (Park 2020; Tyce 2020).

M-Pesa as a global innovation

To consider M-Pesa and its IPR, it is important to step back to consider a global picture. As shown in Figure 2, Safaricom, the Kenyan operator of M-Pesa, is part of a global corporate group of firms under the umbrella of the UK-based Vodafone Group. The Vodafone Group has (or has had) stakes in a range of mobile operators across the globe including in Europe, India, Turkey, Australia and sub-Saharan Africa. From the IPO of Safaricom in 2008 until 2017, Safaricom was directly controlled, while a 2017 group

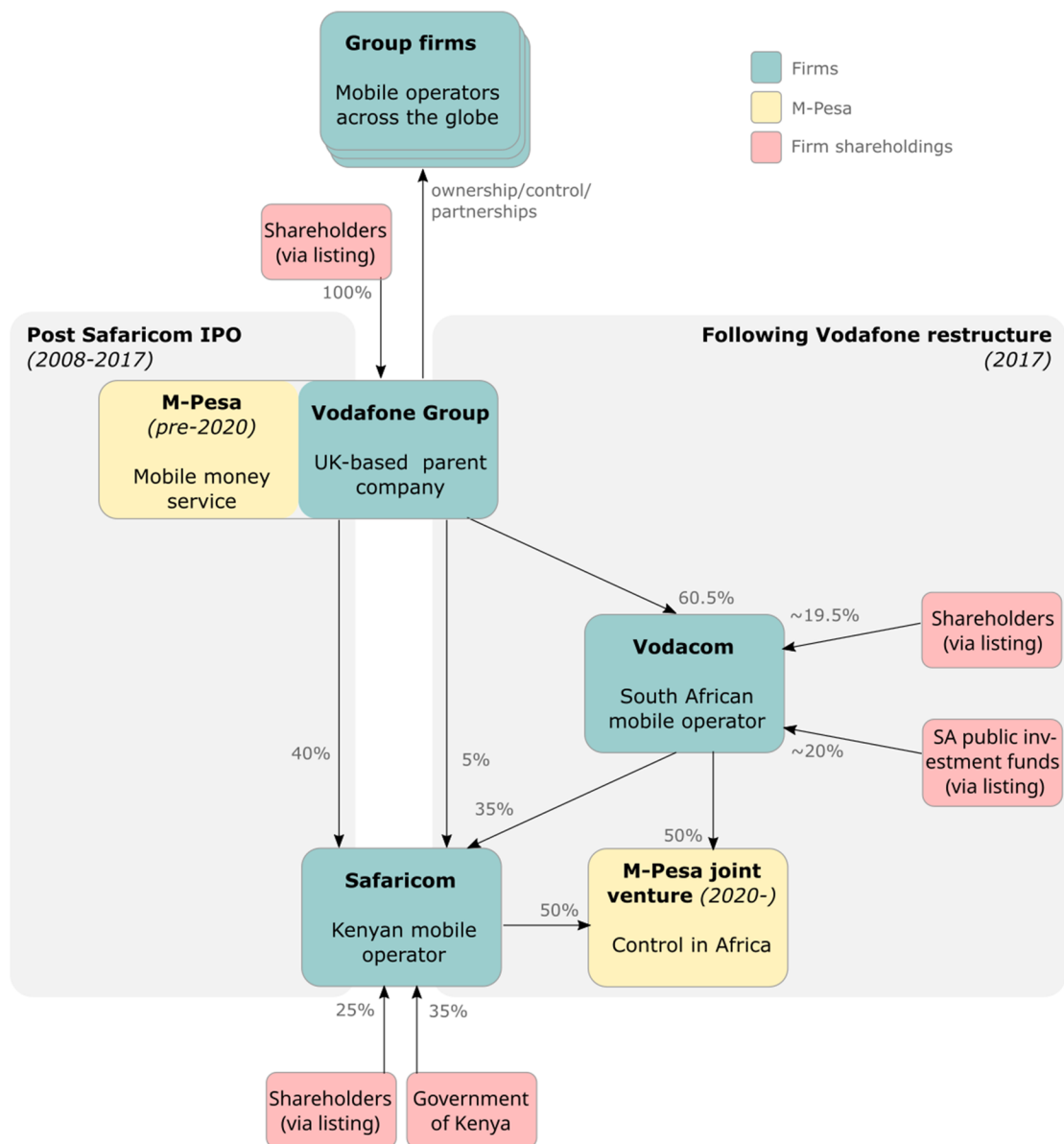


Figure 2. Ownership structure of Safaricom (operator of M-Pesa in Kenya).

Source: Adapted from Vodafone Group, Vodacom, Safaricom annual reports and company presentations.

restructure meant it was indirectly controlled *via*. South African subsidiary Vodacom.

Highlighting this corporate structure is important because, with the success of M-Pesa, the service has expanded from Kenya and become incorporated into the Vodafone Group strategy to leverage mobile money globally. This has included similar mobile money implementations within its subsidiaries including Egypt, Ghana, DR Congo, Tanzania, Mozambique, Lesotho and India. The Kenyan implementation remains the most successful and largest mobile money market for Vodafone Group (23% of all m-money customers were from Safaricom in 2019). Even if other countries have not matched the successes of

Kenya,² the service remains an important source of revenue for Vodafone. In 2019, 12% of all Vodafone Group's service revenue in emerging markets came from M-Pesa (Vodafone Group 2019).

While the operators of subsidiary telecom firms are connected to the parent Vodafone Group, it is important to highlight the potential for different motives amongst subsidiaries. As shown in Figure 2, Vodafone Group is fully listed in the UK stock market. It has a controlling stake in Safaricom, but this is only a minority stake. Other shareholders in Kenya include the government as well as local shareholders (who are said to have been captured by elite groups in Kenya) (Tyce 2020). With varying shareholders,

Table 2. Major intellectual property rights (IPR) related to M-Pesa.

	Details	IPR coverage	IPR details
Branding	M-Pesa marks and logos used in Kenya and globally within service and for branding.	Trademarks	2006–2008: M-Pesa brand and “phone logo” in Kenya filed by Safaricom; 2010–2012: M-Pesa brand in Tanzania filed by Vodafone Group; 2010–2012: M-Pesa brand and “smartphone logo” in EU filed by Vodafone Group.
Software	Mainly the “back-end” server software which enables mobile money systems. Software for phone applications.	Propriety-developed software automatically covered by copyright. Specific novelty, algorithms within software operation are covered by patents (see below).	Software solely owned by Vodafone Group. Software development and hosting outsourced (see discussion in “Leveraging M-Pesa IPR” section).
Novel ideas	Novel aspects of the M-Pesa service in operation. Novel service management aspects, extensions in use and processes.	Set of patents filed related to M-Pesa from 2005–2015: 7 granted, 3 not granted (see Table 4).	Filed mainly by UK actors, working for Vodafone Group (see discussion in “Expansion and innovation” section).

Source: Authors analysis of company reports and IPR databases.

agendas between different arms of the broader corporate group can diverge.

M-Pesa IPR

The major IPR related to M-Pesa includes trademarks, patents and the copyright embedded in the software as shown in Table 2 and discussed in the subsequent sections.

The invention of M-Pesa

As outlined in the previous section, the origins of M-Pesa were in a project involving the UK development agency DFID. Of the £1.96 m, the donor contributed £970,000 and Vodafone Group £990,000. DFID’s monies came from its Financial Deepening Challenge Fund (FDCF), which supported commercial firms to become more involved in development – typically such funds in the development sector have focussed on enabling commercial firms to overcome “risky” markets in the Global South and exploiting “latent demand” for products (Brain, Gulrajani, and Mitchell 2014). With comparatively limited funds and funds available as discrete short-term projects, such Challenge Funds were designed with little expectation of innovation (Pompa 2013). With innovation being a peripheral aspect, approaches to dealing with intellectual property were rarely well thought out. Thus, when a specific project became innovative (as in the case of M-Pesa), there was a lack of clear processes to deal with these issues.

Internal reports and reviews highlight how such issues around IPR arose in the DFID/FDCF fund. In the mid-term review, IPR is brought up as a significant problem given several projects involved ICT implementations.

The implementation of approved projects to-date has been impeded by contractual issues (primarily arising

over IPR) ... Intellectual property rights were also not adequately considered in the design process. As many proposals involve customised ICT development, this poses a serious problem (Ebony Consulting 2003, 9)

The long-standing principle for donor-supported projects in the UK is that IPR should be shared. Firms should “give DFID a license to use the project material this would include the right to publish, copy and distribute all material include written material and software produced under a FDCF grant” (Ebony Consulting 2003, 28). This, however, proved problematic in the FDCF where some private firms argued that “signing an agreement that potentially gave DFID rights to their IPR was a problem” (Irwin and Porteous 2005, 9). In the case of the FDCF, workarounds appear to have been made:

DFID thereafter explained its position which is to require IPR co-ownership only on software developments made under grant funding. That means that if the initial software was acquired before the grant, the institution will retain exclusive IPR (Ebony Consulting 2003, 28).

Overall, in reflecting on such experiences, the review of the FDCF suggests that the best way forward would be to:

... avoid funding the development of new technology ... it avoids getting into debates about who ultimately owns the IPR (Irwin and Porteous 2005, 44).

Other sources of tensions around IPR in the origins of M-Pesa can also be observed in the pilot study and early life of the service. Firstly, tensions about who controls the service amongst different entities of the broader Vodafone Group (see Figure 2). In the 2003 proposals, for example, it is specifically highlighted that there were three financial contributors – Vodafone Group (£390,000), Safaricom (£300,000), and Vodacom Tanzania (£300,000) (DFID 2002). Secondly,

the observation that key contributors to M-Pesa have come from beyond the designers and developers in Vodafone and Safaricom. Principal amongst these contributors were trial users/agents in the pilot project who have been argued to play a crucial, and active role in shaping the M-Pesa service into what it is today through their appropriation of technology during the pilot (Foster and Heeks 2013a; Hughes and Lonie 2007). There are also well-documented precursor innovations that influenced the form of the pilot, including by informal entrepreneurs within East Africa where mobile transfers were well-known before pilot studies.³ Other M-money services such as SmartMoney (created in 2000), G-Cash (created in 2004) (Philippines), and CelPay (created in 2002) (Zambia) also pre-date M-Pesa and these were important precursors to M-Pesa (Porteous 2006).

Although one can, therefore, observe diverse actors involved in the emergence of M-Pesa across multiple domains and scales, this was not reflected in the IPR itself. From its launch in 2007, the IPR (software, brands, patents) was wholly controlled by

the Vodafone Group in the UK, as shown in Figure 3(a).⁴ This had consequences. To use the M-Pesa service, Safaricom in Kenya paid an agreed annual fee to Vodafone Group UK. This began in 2007 as 32.5% of M-Pesa revenue but has reduced over time to 10%, 5%, and finally 2%. This was positioned as a “service fee” that allowed Safaricom to use the M-Pesa service (and permitted the use of IPR assets).

The financial impact of this “service fee” has been significant transfers from Safaricom to Vodafone Group. Figure 4 and Table 3 provide an estimate of the value of service fees, which reached a peak of around €26m in 2015. Over the period 2010–2019, transfers between Safaricom and Vodafone Group totaled around €170m. Running such a service has costs in terms of maintaining and upgrading systems and software. Nevertheless, these figures represent significant transfers between Kenya and the UK. The extent of these service fees might also be surprising given the earlier outlined origins of M-Pesa as part of a development project.

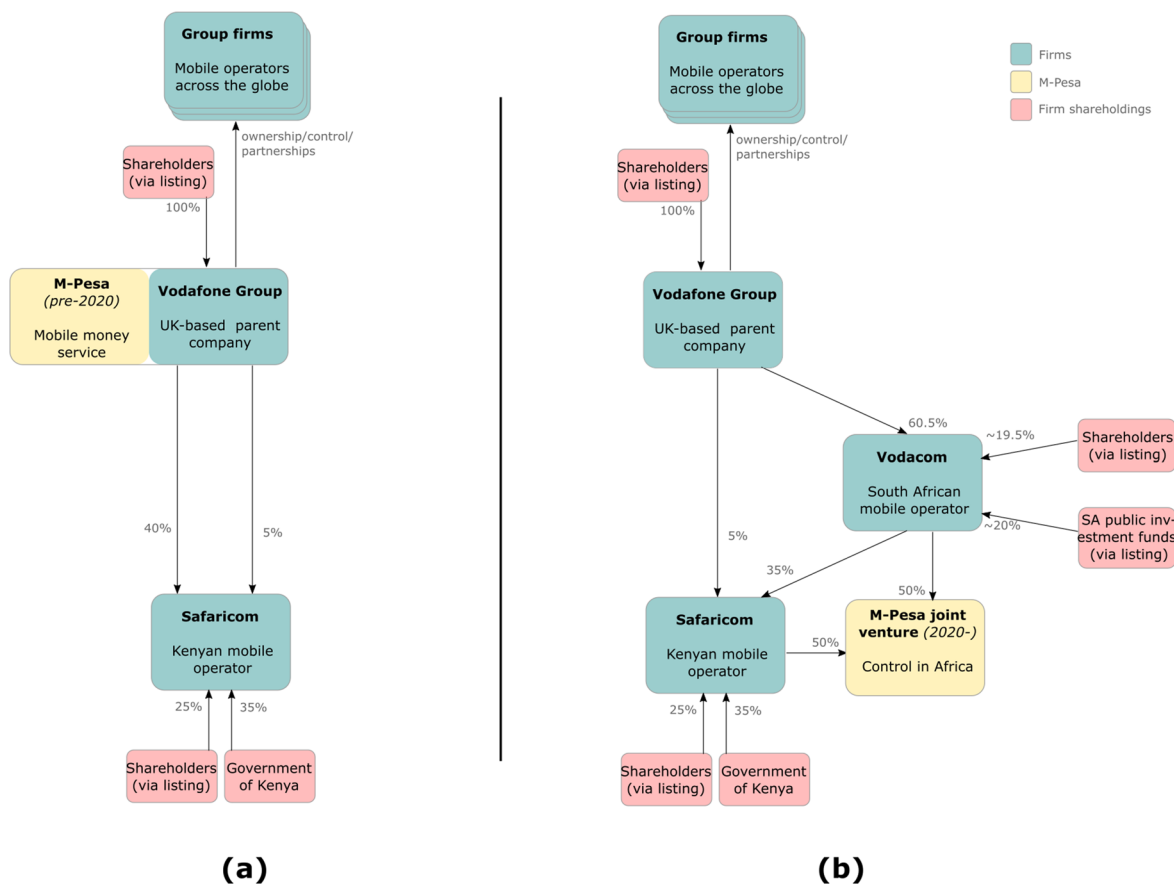


Figure 3. Position of M-Pesa IPR within ownership structure: (a) during early years of M-Pesa (b) following 2020 African joint venture.

Source: Adapted from Vodafone Group, Vodacom, and Safaricom annual reports and company presentations.

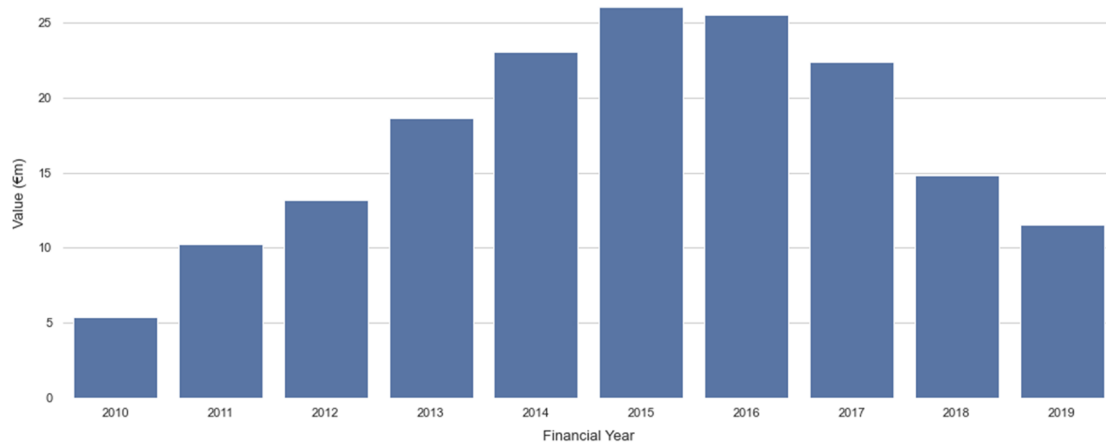


Figure 4. Estimated service payments for M-Pesa.

Note: See Table 3 for details.

Source: Author's calculations based on company reports.

Table 3. M-Pesa Revenues and service fees.

Financial year	10	11	12	13	14	15	16	17	18	19
M-Pesa revenue (€m)	72.0	99.6	152.5	198.7	222.4	327.7	359.4	501.4	505.0	663.2
M-Pesa service fee (%)	10	10	10	10	10	10	10/5*	5	2	2
Estimated service payment (€m)	5.3	10.2	13.2	18.6	23.1	26.1	25.6	22.4	14.9	11.6

Note: *Fee moves from 10% to 5% on August 1, 2015.

Source: Author's calculations based upon company reports. See Appendix for a detailed discussion of calculations.

Expansion and innovation

Innovation dynamics and bottlenecks

Following the pilot, M-Pesa's growth and expansion align with the notion that innovation is not a singular, "one-shot" act. Rather growth should be seen as including incremental improvements that were central to success at scale (Foster and Heeks 2013b). For example, the refinement of core business models and agent business processes was important as the service rapidly scaled. Additional service functionality, such as bank savings, microenterprise integration, and store payments emerged as part of the service expansion and to ensure M-Pesa was able to compete against early mobile money competition (Foster and Heeks 2013a).

In these contexts of incremental innovation, control of IPR affected the trajectories of the service. In interviews with those involved in M-Pesa in Kenya, there was ongoing frustration about the relatively slow and limited upgrading of new services and features, with Safaricom tending to prefer simple integration of new actors over radical changes. For example, during 2012, the network was marked by repeated dropouts, quality and security issues where Safaricom was perceived to be slow to respond by agents and customers

(Foster 2014). Later in 2016, frustration came from technical developers and third-party firms due to the perceived slowness in releasing an application programmable interface (API) to support commercial integration of M-Pesa, and the quality of the smartphone apps was limiting for more advanced M-Pesa use within retail.

These bottlenecks have often been put down to Safaricom being a market monopoly in the press, which does not feel pressure to push radical innovation. However, in interviews, a major factor discussed related to control of the server software. Vodafone Group's control made it more difficult for Safaricom to rapidly upgrade. The main goal for Vodafone Group at the time was to optimize M-Pesa across multiple jurisdictions (particularly in the earlier period when mobile money was considered to potentially be a disruptive technology that would be rolled out across many nations). Safaricom's directions were narrower – looking toward maintaining its market dominance and leveraging the service in broader financial applications in Kenya. However, with control of IPR within Vodafone, it became more difficult for Safaricom to lead and/or influence the directions around service and push technical upgrading, particularly in the software.⁵

Patents and innovation

As M-Pesa has expanded, it has been associated with a growing number of patents (listed in Table 4).⁶ Of the 10 patent filings that were found, seven have been granted. All patent inventors were associated with staff working for the wider Vodafone Group (and its contractors) with no Kenyan or even African-based patent holders. In terms of content, a few patents appear to be associated with implemented features and upgrades of the M-Pesa service (e.g. #6, which aligns with the “Fuliza” service launch in 2019 that provides a small overdraft facility for some M-Pesa users). Others appear less connected to current functionality (#3, #9).

Table 4. Patents linked to M-Pesa.

Title and ID	Inventors	Priority date	Area
1. Mobile account management (WO2007020394)	From Vodafone, UK-based & UK-based M-Pesa software consultants	2005	World
2. Mobile transactions (WO2012052787A1)	From Vodafone, UK-based	2010	World
3. Delivery system and method (WO2012168735A1)	From Vodafone, UK-based	2011	World
4. Cellular airtime management (US9462138B2)	From Vodafone, UK-based	2012	USA, Africa
5. Mobile savings account (AP2013007002A0)	From Vodafone, UK-based	2012	Africa
6. Mobile money transfer (AP2013007001A0)	From Vodafone, UK-based	2012	Africa
7. Optimizing financial transactions network flow (US20150254651A1)	From Vodafone, EU & US-based	2014	Not granted
8. Determining multiple users of a network enabled device (US9301126B2)	From Vodafone, EU & US-based	2014	USA
9. Mobile transaction validation (GB2534863A)	From Vodafone, UK-based	2015	Not granted
10. Digital currency conversion (GB2537683A)	From Vodafone, UK-based	2015	Not granted

Source: Authors analysis based on patent search.

Three aspects of the filings were further examined in line with the literature discussion on IPR: (a) the sources of novelty for the patents, (b) the basis of novelty claims, and (c) the type/scope of the patents.

Source of novelty: Three sources underlie the novelty of claims. A first cluster formalizes mobile/mobile money practices in software that had already emerged informally in Kenya (#1, #4, #5, #6). A second solve specific technical problems emerging as M-Pesa has expanded as a service (#2, #8, #9). The third describes new or future features (#3, #7, #10).

The first category is the most controversial in terms of the literature discussion on IPR. A number of the patents align with common practices occurring amongst users of mobile, or mobile money in Kenya. The patents then look to formalize these practices, often by describing potential software systems which would, for the most part, support these practices.

An example of this is the principal 2005 patent (#1) whose “claims”⁷ cover a software system to manage payments for MFIs (micro-finance institutions), where MFI transfers might be done through mobile messaging. The presence (as outlined in the previous section) of precursor services such as G-Cash and Celpay, as well as practices in Kenya within MFIs and informal ROSCAs (rotating savings and credit associations), suggest that much of the novelty in this patent links to activities that were already emerging in Kenya.

The basis for novelty claim: In most of the patents, the major novelty claim is the *presence or application of mobile money*. A group savings patent (#5), for example, has parallels to informal approaches to group savings in Kenya. Its basis for novelty rests on the system proposed. Rather than saving groups distributing cash amongst group members, they would instead use systems that integrate automated mobile money and/or mobile airtime conversions. In this sense, many of the patents might be classed as “software patents”. They describe software and aspects of mobile money operations and processes.

Scope of patents: In terms of scope, at least if one takes them at their broadest claims, the patent defines extremely broad fields of novelty.⁸ The initial mobile account patent (#1), for example, defines a broad technical area which might impinge on many MFI systems that integrate mobile money. Similarly, the patent that outlines a technique for the management of payments over airtime (#4), could suggest patent infringement in a growing number of payroll or non-profit payment systems that incorporate mobile money payments. However, there is little evidence that patents have been used in significant external litigation, thus these arguments remain only hypothetical at present.⁹

Leveraging M-Pesa IPR

Recent activities around M-Pesa have moved into new territories. These discussions reinforce the previous arguments that IPR is key to control of the service, but they go beyond this. They highlight some ways

that corporate activities and power within the Vodafone structure relate to IPR.

Controlling M-Pesa software upgrades

The “back end” software enables the operation of the M-Pesa system. During the initial DFID pilot, this was contracted to a UK-based software firm, Sagentia. As mobile money interactions did not align with existing banking systems, proprietary software was written (Wooder and Baker 2012). Given the interactive processes of innovation during the M-Pesa pilot and the early stage of the rollout, the software inevitably needed to be adapted over time, dependent on the evolving business proposition and regulatory requirements (Wooder and Baker 2012). As the service grew, the responsibility for the software was directed to IBM in 2009.

Following earlier struggles with the software, significant changes were made during the M-Pesa “G2 project” (concluded in 2015). This was a cross-country software project involving Vodafone, Safaricom, IBM with Huawei as the main software development partner. The aim was to upgrade the software efficiency given its growing use and publicized delays. Most symbolically, the upgrade was accompanied by a first celebration of “bringing M-Pesa” home, when M-Pesa servers and software were relocated from Germany to Kenya (even as the IPR remained with Vodafone Group).

It was only later that questions surrounding the G2 project and IPR came to light. Following broader accusations of corruption in Safaricom, auditors KPMG were commissioned to undertake a forensic investigation of recent contracts. The report was leaked to the Kenyan press in 2016 (KPMG 2016). It reveals the cost to Safaricom for the M-Pesa upgrade to be \$12.5m. This involved a disputed procurement process with KPMG commenting that “based on the value of other proposals ... the cost agreed with Huawei is relatively high” (KPMG 2016, 111). Crucially, concerns were also raised about the IPR ownership associated with this software. KPMG reported that:

... although the cost of this project was borne largely by Safaricom, it does not retain exclusive use of intellectual property ... similar mobile money projects can be rolled out in the territories [other Vodafone subsidiaries] without any benefits to Safaricom (KPMG 2016, 110).

According to the report, Safaricom was made to pay the full cost of the upgrade project (due to the reported dissatisfaction of Kenyan managers). Yet

there was no extended control of the IPR, which remained solely owned by the Vodafone Group.

IPR in the 2020 joint venture

Control of the M-Pesa (and its IPR) moved within the Vodafone Group corporate structure in 2020. It was moved to an African joint venture (JV) between Safaricom and South African subsidiary Vodacom, as shown in Figure 2(b). The two firms each paid 1,073 million KSh (£7.02m) to Vodafone Group for the IPR (Phillip 2020).

This change was part of a broader corporate restructuring of the Vodafone Group. Official explanations for this reshuffle mention the regional simplification of structures (Vodafone Group 2019), but the new structure also aligns better with a future sale of Vodafone’s African/East-African assets (Fildes 2017). By the late 2010s, the expected expansion and disruption of mobile money was more modest than earlier anticipated. With Vodafone also selling off several other subsidiary operators (such as in India), the main source of mobile money revenue is now Africa (and mainly in East Africa). Therefore, placing the M-Pesa IPR within this part of the company would be important to the valuation and/or viability of an entity as part of a future corporate sale.

These changes and enhanced regional control of the M-Pesa IPR appear to be overcoming some of the previous bottlenecks. Associated with JV were strong articulations by African managers that this would lead to a renewal in the development of M-Pesa.

This JV... will be investing in the next generation M-Pesa platform, an open, intelligent digital platform designed for smartphones. The JV will also invest in best-in-class anti-money laundering and cyber security capabilities to defend M-Pesa from these growing threats (Vodacom 2019, 35)

Indeed, following this restructure it has been notable that a new M-Pesa application was launched incorporating “fintech-style” features including enhancements for tracking payments, bills, people and with improved QR code use.

Discussion

Monopolization/assetization

Examination of M-Pesa in detail highlights how this scaled innovation, even with its initial focus in East Africa, has become global. As it has become a global service, it has become entwined with the strategies of

the multinational parent firm, with goals to maximize its global profits. This has been accompanied by expansionary IPR within the organization. The discussion, therefore, is in line with the ideas of global digital firms building IPR to support their intangible assets. There is a presence of a growing range of IPR, and this is defined quite broadly, as illustrated in the case of patent assets.

M-Pesa's development has involved innovative activities and development by staff including firm managers and software developers which should not be underplayed. Yet, the extensive research of M-Pesa elucidates the reality that a diversity of actors contributed to innovation – development organization, policymakers, agents, entrepreneurs, and service users, including more marginal actors in Kenya. However, it was the managers and technologists in the UK who captured knowledge and were able to write the software code. They are the ones who have been able to stake claims to novelty and this is reflected in the ownership of the IPR. This then highlights dimensions of appropriation of knowledge by IPR in this case. Local “grassroots innovations” around the use of technologies are commodified and extended by corporate actors for profit.

There is less evidence, at least in this case, of the more critical views that IPR is “closing down the commons” or globally monopolizing innovation. Certainly some of the M-Pesa IPR, on paper, might overlap, limit, and/or monetize innovations used elsewhere in the sector. But this is not happening at present. IPR has only played a limited and defensive role with little impact on the broader sector.

The power of IPR, however, is crucial to the shaping of profits and assetization internally to the corporate structure. At its simplest, it is worth restating that the entity (corporate actor) that controls the IPR portfolio is better able to capture profits from innovation, no matter who has been involved in the actual innovation. IPR ownership, for example, is core to the justification for significant transfers of funds moving from Kenya to the UK in the form of service fees (as well as the additional cost of software contracts and the 2020 purchase of the IPR). The 2020 joint venture shows how the M-Pesa IPR later became an important *asset* of the corporate entity. This is one part of the capital stock that forms the basis of the valuation of a firm being bought and sold within corporate markets.

Innovation

The extensive research done over the history of M-Pesa is valuable in that it has documented the role that different actors play in innovation. In line with

contemporary innovation literature, this suggests continuous innovation with diverse contributors.

A closer analysis of the tensions between control within IPR and diverse innovation can be made in this case. Previous accounts examining intellectual capital have focused more on ecosystem relations (Rikap 2022) and value chains (Durand and Milberg 2020), taking them to be the arenas where control is contested. This article supplements this by highlighting the role of inter-firm relations, given the complexity of parent-subsidiary relations in the globalized economy. The different corporate entities discussed are interlinked but can have different agendas and directions around innovation. In this case, IPR is an important part of how innovation is controlled within these corporate relationships. For those operating the service in Kenya, important decisions about directions of innovation have often been defined by the parent company in the UK. This should not be surprising given the parent-subsidiary firm relations, but it is important to emphasize that corporate entities can have diverging goals given the different firm shareholders and stakeholders in the UK and Kenya.

Breaking down further, IPR is made up of different assets (including software, patents, branding). It has been very difficult to separate the role of different assets in this case, given the nature of the M-Pesa service. The software has been particularly important, as a central site of control of how the service operates, but this control is only possible because ownership is supported by the wider IPR assets. Further literature on intellectual capital has focused on other assets such as data, digital intelligence (i.e., analytics from data) and machine learning algorithms. These forms of capital are said to also support firm control and may require less protection within IPR (i.e., data hoarding and proprietary algorithms). There is some evidence of similar resources being used within the M-Pesa service, such as data (Acimovic et al. 2022), but an examination of these more novel resources and their link to IPR is beyond the scope of the article and an important part of future research agendas.

Structural inequality

The present IPR system is poorly set up to deal with the nature of modern innovation, and IPR regimes might further enhance structural inequalities. Several aspects of IPR regimes exacerbate these issues. The way that IPR is associated with a single moment of invention and its attribution to individuals can be challenging. The fact that at an institutional level, a relatively broader set of IPR is also being granted (for

software and business model patents) leads to the ability for larger IPR portfolios and control for those who can stake claims on innovation.

Recent resistance to even modest reforms in IPR, for example, linked to urgent COVID-19 patents, has illustrated the challenges in reforming present IPR regimes. However, it is important to outline visions of IPR that could support improved inclusion. For the diverse direction of innovation, public or socially-orientated bodies need to be more attuned to IPR issues to ensure fairness in IPR. This was, on paper, the case for the UK development agency DFID – a large public organization that had established rules on IPR. However, in implementation, a lack of attention to intellectual property during the pilot has been highly problematic in this respect. This suggested that building capacity and standardized approaches to protect against the capture of IPR should become a more central concern. This would be important, particularly in the digital economy, where firms often begin at a small scale and can rapidly grow to a national or global level.

This case also poses insights into IPR debates on the appropriation of local knowledge or practices. In the digital economy, appropriation of knowledge is present and relates to the way that local practices that emerge within communities or small firms (often involving frugal innovations) can be captured and defined more formally within software patents and lead to potentially profitable services. Such dynamics might push a reconsideration of whether and how such knowledge should be considered and rewarded within IPR regimes.

Conclusion

An analysis of IPR has been used to expand an understanding of aspects of intellectual capital and innovation in the digital economy. The literature review outlined key tenets of IPR debates related to the structural core-periphery relationships, monopolization in the digital economy, and the diversity of innovation and actors involved in IP creation.

Within the context of a globalization digital economy, this literature highlights the need for expanded research. It is important to link more closely between the seemingly contrasting directions of monopolization of IPR vs. the growing diversity of innovation as a distributed process. With the globally expanding digital economy, a lack of empirical analysis requires renewed attention.

In the case of M-Pesa, IPR becomes a key mechanism for control of the direction of innovation,

including the tensions around IPR within the pilot, expansion of IPR within the corporate structure, and use of IPR for control and profit. In effect, although innovation has been diverse and significant, IPR has been claimed and then used as a basis for control.

Reflecting on these discussions, the article has highlighted key challenges embedded in regimes of IPR. Most notably, the way that diverse innovators are underplayed within IPR, fosters corporate control. Consequently, there is a need for broader knowledge and capability in dealing with IPR among all the stakeholders.

These findings suggest that further research is needed to think more carefully about best practices and policies around IPR. Empirical research which opens up further cases of global digital innovation would be valuable. More technical work can also be done to consider how those involved in socially orientated innovations and local practices can be better embedded within IPR systems.

Such work is crucial given that the digital economy is becoming increasingly important globally. As this case shows, there are risks that IPR leads to control of innovation and controversial South-North transfers of resources. The different challenges of IPR regimes discussed highlight the need for further research on innovation through the prism of IPR. These regimes are arguably not fit for purpose in the emerging era of digital capitalism.

Notes

1. Given this article's position, it is appropriate for the author to acknowledge the work has been inspired by a vibrant discussion amongst interviewees and online centered on the debate "is M-Pesa a Kenyan innovation?". Notably, those within Kenyan media and blogosphere have made interesting interventions to this discussion which are relevant to this analysis. This paper therefore acknowledges discussions by Alex Kuria, Tefo Mohapi, Brian Gachichio, Cyprian Nyakundi, Kenya West and Nairobi Law Monthly.
2. For instance, in South Africa, M-Pesa was relatively unsuccessful. Introduced in 2010 shortly after Kenya it never reached a critical mass of users and was withdrawn in 2016.
3. For example, in research undertaken between 2004 and 2006 Nokia ethnographer Jan Chipchase outlined a method of cash transfer referred to as Senté occurring amongst mobile kiosk operators in Uganda (Chipchase and Tulusan 2006).
4. M-Pesa IPR is framed as follows within company documentation "All intellectual property rights in M-PESA belongs to Vodafone and are licensed to Safaricom under a trademark licence agreement" (Safaricom 2008 p.136).

5. Indeed, when this finally became imperative during the 2015 upgrade in Kenya, Safaricom was subject to a significant fee to the group for this.
6. A patent search was made for patents that included the terms “mobile money”, “mobile finance” or “M-Pesa”. As this was a relatively small set of patents, it was possible to manually inspect authors and patent outlines to include those relevant to the case.
7. The formal and legalistic way that the scope of the patent is defined is through the “claims” of a patent.
8. A broad definition would be expected for professionally written patents, with the goal of the initial *independent* claims to define their novelty as generically as possible (to widen their scope of possible claims). Subsequent *dependent* claims within patent then provide more detailed contexts. This style allows patents to cover as broad an area as possible, where dependent claims allow more refined claims to protect more specific embodiments, should the broader claims be found to be invalid during litigation.
9. In Kenya, court actions have occurred where others have argued they invented aspects of M-Pesa. Filing dates on earlier patents have been defensively used to allow Vodafone/Safaricom to claim prior art in such cases.

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Appendix: Notes on calculations of M-Pesa service fees

Details of the service agreement(s) (or service fees paid) are not in the public domain. However, corporate documentation and financial reports from Vodafone Group highlight the method by which M-Pesa service costs are calculated. Some details and limitations of this calculation are highlighted in this Appendix. Service fees are calculated as follows:

1. a percentage of M-Pesa revenue (on a sliding scale as shown in Table 3).

The service fees for a financial year have been calculated based on quarterly revenue splits, delayed by one quarter. This is multiplied by the service commission. Due to a lack of quarterly data about M-Pesa revenue, this figure is estimated by dividing annual data equally into quarters. This simple approach may misrepresent quarters where there is notable growth or decline.

2. an additional component concerning “indirect revenue gains of Safaricom as a result of savings in airtime commission costs”.

This concerns the scenario where customers use M-Pesa to top-up mobile airtime. In this scenario, Safaricom saves on distribution costs and commission payments to local mobile vendors. It is not possible to calculate this based on publicly available documentation.

In sum, due to these limitations, Table 3 should be seen as an estimate of the M-Pesa service fee. For triangulation, in several years data has been released within corporate reports of the exact fee paid for M-Pesa services (for 2011 and 2016–2018). This suggests the above approach is a good estimate, but typically an underestimate by up to €3 m.